Computational Methods in Macroeconomics

Lecture 3

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Heterogeneous Agent Models
Bewley Models

Bewley Models

- So far, we have focused on a partial-equilibrium (fixed prices) setups with a representative agent
- We will now analyze an economy with heterogeneous agents
- To illustrate the techniques, let's consider the simplest Bewley economy

Bewley Models

- We have solved the income fluctuation problem in partial equilibrium, fixed R
- We now close the economy by looking at the equilibrium on the goods and asset markets
- We consider the same saving-consumption problem as before, but now we consider a measure one of ex-ante identical households
- There is no aggregate risk (yet!) $\rightarrow$ Aggregate income is equal to 1
- Given $R(.)$, a household with current assets a and income y, solves this problem:

$$V(a,y) = \max_{c,a'} \; u(c) + \beta \mathbb{E} V(a', y')$$

$$\text{s.t.} \quad c + a' = R(.)a + y$$

$$\log y' = \rho \log y + \sigma \epsilon'$$

$$a' \geq -\phi \;\; \text{(borrowing constraint)}$$

- Let $\Omega(a,y)$ denote the distribution of agents over states
- Our first task is to characterize how agents move across states over time $\rightarrow$ we need to obtain next period's distribution, given the current one and policies.

Bewley Models

- As before, we create a grid for a, $\underbrace{\{a_1, ...a_n\}}_{\mathcal{A}}$ and y, $\underbrace{\{y_1, ...y_m\}}_{\mathcal{Y}}$
  - We also need to discretize the ϵ shock
  - Only to simplify notation, let me assume that we are using Tauchen's method (no interpolations across y')
  - Let $\pi(y, y')$ denote the transition probability from y to y' (y' is on grid)
- Let $a'(a, y)$ and $c(a, y)$ denote each agent's policy functions
  - For a given R, we know how to solve the previous problem and find those policies...
  - The extra layer of difficulty now is that R is endogenous... and it depends on the Ω distribution
- Define $Q((a, y), A, Y)$ as the probability that an agent with current state (a, y) transits to the subset $A \times Y$, with $A \in \mathcal{A}$ and $Y \in \mathcal{Y}$

$$Q((a, y), A \times Y) = \mathbb{I}_{\{a'(a,y) \in A\}} \sum_{y' \in Y} \pi(y, y')$$

- where $\mathbb{I}_{\{a'(a,y) \in A\}}$ is an indicator function that equals one if $a'(a, y) \in A$.

Bewley Models

- The law of motion for the distribution of agents is then:

$$\Omega'(A \times Y) = \int Q\Big((a,y), A \times Y\Big) \times \Omega(a,y)\, da\, dy$$

- Let me re-write the problem more formally now, to take into consideration the dependence between R and Ω:

$$\begin{aligned} V(a,y,\Omega) = \max_{c,a'} \; & u(c) + \beta \mathbb{E} V\left(a', y', \Omega'\right) \\ \text{s.t.} \quad & c + a' = R(\Omega)\, a + y \\ & \log y' = \rho \log y + \sigma \epsilon' \\ & a' \geq -\phi \;\; \text{(borrowing constraint)} \\ & \Omega'(A \times Y) = \int Q\Big((a,y), A \times Y\Big) \times \Omega(a,y)\, da\, dy \end{aligned}$$

Bewley Models - Stationary Recursive Competitive Equilibrium

- A stationary RCE is a value function $V(.)$, policy functions $a'(.)$ and $c(.)$, prices R, and a stationary distribution Ω, such that:
  - Given prices R, the policy functions $a'(.)$ and $c(.)$ solve the households' problem and $V(.)$ is the associated value function
  - The asset market clears $0 = \int a'(a,y)\Omega(a,y)\mathrm{d}a\mathrm{d}y$ (bonds are in zero net supply)
  - The goods market clears; redundant due to Walras law
  - For all $A \times Y$, the stationary (or invariant) distribution satisfies:

$$\Omega\left(A \times Y\right) = \underbrace{\int Q\left((a,y), A \times Y\right) \Omega\left(a,y\right) \mathrm{d}a\,\mathrm{d}y}_{\equiv \Omega'(A \times Y)}$$

Bewley Models - Stationary RCE

- For general conditions under which an equilibrium exits and is unique $\rightarrow$ see Stokey-Lucas-Prescott, Recursive Methods in Economic Dynamics
- You saw with Gaston these properties...
  - **Feller property**: stochastic law of motion is "continuous" with respect to the current state
  - **Compactness / bounded state space:** the set of possible states is closed and bounded (income process, for instance, does not drift off to infinity)

Bewley Models - Stationary RCE - Algorithm

- Create grids for a and y
  - (suppose we approximate $\{y'\}$ process using Tauchen's method)
- Fix an **initial guess** for the interest rate $R^0 < \frac{1}{\beta}$ (remember that we need $\beta R < 1$)
- Given R^0, we can solve the household dynamic programming problem
  - VFI, policy function iteration, endogenous grid method...
- Given the policy function $a'(a, y)$, need to find the invariant distribution Ω
- For this, we construct the transition function $Q(.)$, and by successive iteration over the LOM

$$\Omega(A \times Y) = \int Q((a, y), A \times Y)\, \Omega(a, y)\, da\, dy$$

we obtain the fixed point distribution $\Omega^\star$

Bewley Models - Stationary RCE - Algorithm

- There are many ways to implement the last step: finding the stationary distribution $\Omega^\star$
  1. **Stochastic (Monte Carlo) simulations** (sampling error; can be slow)
  2. **Non-stochastic simulations (Young, 2010)** (faster; generalizes well to richer environments)
  3. **Solve directly for the invariant distribution** using eigenvectors
- Once we have $\Omega^\star$, we check whether the asset market clears:

$$0 = \underbrace{\int a'(a,y)\Omega^\star(a,y)\mathrm{d}a\mathrm{d}y}_{\bar{a}}$$

- If $\bar{a} > 0$, there is an **excess supply** of assets $\rightarrow$ in the next guess we should **lower** R
- If $\bar{a} < 0$, there is an **excess demand** of assets $\rightarrow$ in the next guess we should **increase** R
- We then update our guess for R and re-start the algorithm
- We stop when: $|R^{n+1} - R^n| <$ convergence criterion

Bewley Models - Monte Carlo Simulations

- The most direct approach is to simulate a large number of households over time
- Start from an initial cross-sectional distribution of agents $\Omega_0(a, y)$
- For each household i and period t:
  - observe current state $(a_{i,t}, y_{i,t})$
  - use the policy rule to compute next-period assets:

$$a_{i,t+1} = a'(a_{i,t}, y_{i,t})$$

  - draw the income shock and update $y_{i,t+1}$
- Repeating this across households gives a simulated cross-sectional distribution at date $t+1$
- For a very large T, the empirical distribution (histogram), Ω_T, approximates the invariant distribution $\Omega^\star$

Bewley Models - Monte Carlo Simulations

- Once the simulation has converged (large T), we approximate the stationary distribution using the histogram of simulated agents over the (a, y) grid
- Aggregate asset holdings are then computed as:

$$\bar{a} \approx \frac{1}{N} \sum_{i=1}^{N} a'(a_i, y_i)$$

- Advantage:
  - conceptually simple
- Disadvantages:
  - subject to sampling error; problematic when solving for prices and market clearing
  - can require many agents and many periods
  - convergence can be slow
- This is why non-stochastic methods are often preferable...

Bewley Models - Solving for the Invariant Distribution Directly

- Suppose we have n points for the a grid and m points for the y grid
- We stack (**vectorize**) the state space (a, y) into a vector of length $n \times m$

$$X = \begin{bmatrix} a_1 & y_1 \\ a_2 & y_1 \\ \dots & \\ a_n & y_m \end{bmatrix}$$

- Compute the transition matrix $Q_{(n\times m)\times(n\times m)}$, where each element $Q_{(i,j),(k,\ell)}$ is defined as:

$$Q_{(i,j),(k,\ell)} = Pr\left[(a', y') = (a_k, y_\ell) \mid (a, y) = (a_i, y_j)\right]$$

- In our construction, each row corresponds to a current state (a_i, y_j), each column corresponds to a next-period state (a_k, y_ℓ), and each row sums to one
- Let $\omega^\star$ denote the **vectorized** invariant distribution

Bewley Models - Solving for the Invariant Distribution Directly

Example

Suppose we have 2 asset points and 2 income points.
Then the stationary distribution can be written as a matrix:

$$\Omega^\star = \begin{bmatrix} \Omega^\star(a_1, y_1) & \Omega^\star(a_1, y_2) \\ \Omega^\star(a_2, y_1) & \Omega^\star(a_2, y_2) \end{bmatrix}$$

Vectorizing means stacking the entries of this matrix into one column vector. For example:

$$\omega^\star = \begin{bmatrix} \Omega^\star(a_1, y_1) \\ \Omega^\star(a_2, y_1) \\ \Omega^\star(a_1, y_2) \\ \Omega^\star(a_2, y_2) \end{bmatrix}$$

Bewley Models - Solving for the Invariant Distribution Directly

- The (stacked) stationary distribution satisfies:

$$\omega^{\star}_{n\times m,1} = Q^{T}_{n\times m,n\times m} \quad \omega^{\star}_{n\times m,1}$$

- We can re-write this as:

$$0 = \left(Q^{T} - I\right)\omega^{\star}$$

- Then, $\omega^{\star}$ is the eigenvector associated with the eigenvalue $\lambda = 1$ of Q^{T}
- Since eigenvectors are only defined up to scale, we normalize $\omega^{\star}$ so that its elements add up to one
- After normalization, $\omega^{\star}$ is a valid probability distribution over the grid

Young (2010) Method to Approximate Ω

- Note: We have defined our grid in terms of $\tilde{y} = \log y$. From now on, I will use $\tilde{y}$ to avoid confusion
- **Idea behind this method:** start from a density over a fixed grid and construct tomorrow's density directly from policies; no need to simulate
- Denote the current density by $\Omega_t(a, \tilde{y})$
- For each grid point $(a, \tilde{y})$, locate the policy function $a'(a, \tilde{y})$ within the asset grid
- That is, find the index J such that:

$$a'(a, \tilde{y}) \in [a_J, a_{J+1}]$$

- Similarly, for each grid point $\tilde{y}$ and each discretized shock ϵ', find the index K such that

$$\tilde{y}'(\tilde{y}, \epsilon') = \rho\tilde{y} + \sigma\epsilon' \in [\tilde{y}_K, \tilde{y}_{K+1}]$$

  - this second interpolation step is only needed if we are not using Tauchen

Young (2010) Method to Approximate Ω

- Assign weights to the 2 closest asset grid points using linear interpolation:

$$w_{a_J}(a, \tilde{y}) = \frac{a_{J+1} - a'(a, \tilde{y})}{a_{J+1} - a_J}$$
$$w_{a_{J+1}}(a, \tilde{y}) = 1 - w_{a_J}(a, \tilde{y})$$

  - where $w_{a_J}(a, \tilde{y})$ is the weight on the lower bound

- Similarly, for the exogenous process:

$$w_{\tilde{y}_K}(a, \tilde{y}, \epsilon') = \frac{\tilde{y}_{K+1} - \tilde{y}'(a, \tilde{y}, \epsilon')}{\tilde{y}_{K+1} - \tilde{y}_K}$$
$$w_{\tilde{y}_{K+1}}(a, \tilde{y}, \epsilon') = 1 - w_{\tilde{y}_K}(a, \tilde{y}, \epsilon')$$

  - where $w_{\tilde{y}_K}(a, \tilde{y}, \epsilon')$ is the weight on the lower bound

Young (2010) Method to Approximate Ω

- We then distribute the mass at $(a, \tilde{y})$ across the 4 neighboring grid points:

$$\Omega_{t+1}\left(a_J, \tilde{y}_K\right) = \left[\underbrace{\sum_{i=1}^{N} w_i \, w_{a_J}(a, \tilde{y}) \, w_{\tilde{y}_K}(a, \tilde{y}, \epsilon_i)}_{\equiv Q((a,\tilde{y}),(a_J,\tilde{y}_K))}\right] \Omega_t(a, \tilde{y})$$

$$\Omega_{t+1}\left(a_{J+1}, \tilde{y}_K\right) = \left[\sum_{i=1}^{N} w_i \, w_{a_{J+1}}(a, \tilde{y}) \, w_{\tilde{y}_K}(a, \tilde{y}, \epsilon_i)\right] \Omega_t(a, \tilde{y})$$

$$\Omega_{t+1}\left(a_J, \tilde{y}_{K+1}\right) = \left[\sum_{i=1}^{N} w_i \, w_{a_J}(a, \tilde{y}) \, w_{\tilde{y}_{K+1}}(a, \tilde{y}, \epsilon_i)\right] \Omega_t(a, \tilde{y})$$

$$\Omega_{t+1}\left(a_{J+1}, \tilde{y}_{K+1}\right) = \left[\sum_{i=1}^{N} w_i \, w_{a_{J+1}}(a, \tilde{y}) \, w_{\tilde{y}_{K+1}}(a, \tilde{y}, \epsilon_i)\right] \Omega_t(a, \tilde{y})$$

- where w_i are the weights associated with the discretized shocks ϵ'
- We repeat this for each $(a, \tilde{y})$ point and then add up all contributions to obtain Ω_{t+1}

Young (2010) Method to Approximate Ω

- If we approximate the exogenous process using **Tauchen's** method, then the procedure is simpler
- We only need to interpolate across the endogenous state:

$$w_{a_J}(a, \tilde{y}) = \frac{a_{J+1} - a'(a, \tilde{y})}{a_{J+1} - a_J}$$
$$w_{a_{J+1}}(a, \tilde{y}) = 1 - w_{a_J}(a, \tilde{y})$$

$$\Omega_{t+1}\left(a_J, \tilde{y}'\right) = \left[\underbrace{\sum_{\tilde{y}'} \pi(\tilde{y}' \mid \tilde{y})\, w_{a_J}(a, \tilde{y})}_{\equiv Q((a,\tilde{y}),(a_J,\tilde{y}'))}\right] \Omega_t(a, \tilde{y})$$

$$\Omega_{t+1}\left(a_{J+1}, \tilde{y}'\right) = \left[\sum_{\tilde{y}'} \pi(\tilde{y}' \mid \tilde{y})\, w_{a_{J+1}}(a, \tilde{y})\right] \Omega_t(a, \tilde{y})$$

Young (2010) Method to Approximate Q

- To find the invariant distribution: we iterate on this mapping until Ω_{t+1} and Ω_t are sufficiently close
- Alternatively...
  - Once we have the transition matrix $Q(.)$, we can compute its eigenvector associated with eigenvalue 1 and recover the invariant distribution $\Omega^\star$ directly

Young (2010) Method

- Notice that the distribution matrix Ω and the transition matrix Q are "huge" — Q size is $(n \times m) \times (n \times m)$
  - Most of its entries, however, are zeros
- It is important to work with sparse matrices, particularly when computing eigenvectors

Young (2010) Method

- Example:

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.3 & 0.7 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0.2 & 0.8 \end{bmatrix}$$

- This matrix is sparse because most entries are zero
- Instead of storing all 16 entries, we only store the nonzero ones:

$$A(1,1) = 1,\ A(2,2) = 0.3,\ A(2,3) = 0.7,\ A(3,4) = 1,\ A(4,3) = 0.2,\ A(4,4) = 0.8$$

- This saves memory and makes matrix operations faster when the matrix is very large

Young (2010) Method

- Example:

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.3 & 0.7 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0.2 & 0.8 \end{bmatrix}$$

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- This saves memory and makes matrix operations faster when the matrix is very large
- Let's see Young's method in Julia...

Transitional Dynamics in Bewley Models

Transitional Dynamics in Bewley Models

- So far, we have focused on stationary equilibria.
- Now, we analyze dynamics to an **unexpected (MIT) shock**
  - There is no aggregate uncertainty...
  - ...the probability of an aggregate shock happening is not accounted in the policy functions...
- **Steady-state comparison**: If the shock is permanent, we can start the analysis by computing a stationary RCE for the case before and after the shock.
- **Transition**: More generally, we are interested in transitory shocks $\rightarrow$ to analyze its effects, we need to compute the whole transition
  - The shock will change the individual consumption/saving decision and therefore aggregate prices
  - Those changes will induce dynamics away from the current steady-state
  - **How can we solve this problem?** "Backward+Forward Transition"

Transitional Dynamics in Bewley Models

- Let's introduce additional features in the model...
  - Suppose that government taxes households income and the initial tax rate is $\tau^\star$
  - At time $t = 1$, the tax rate unexpectedly increases for T_τ number of periods. After that, it goes back to $\tau^\star$
- It is useful to re-write our problem using a t sub-index as follows:

$$V_t(a, y) = \max_{c_t, a_{t+1}} \; u(c_t) + \beta \mathbb{E} V_{t+1}(a_{t+1}, y_{t+1})$$

$$\text{s.t.} \quad c_t + a_{t+1} = R_t a + y(1 - \tau_t)$$

$$\log y_{t+1} = \rho \log y + \sigma \epsilon_{t+1}$$

$$a_{t+1} \geq -\phi \;\; \text{(borrowing constraint)}$$

- Next, we will see a **"Shooting Algorithm"** to solve for the transitional dynamics

Transitional Dynamics in Bewley Models - Algorithm

- Compute the initial SS: $V^\star(a,y), c^\star(a,y), a'(a,y), R^\star$ corresponding to $\tau_t = \tau^\star$
  - For this, we apply the same algorithm that we saw above
  - Since the shock is temporary, the economy will revert to this steady state eventually
- Fix the length of the transition $T > T_\tau$. Set T large enough just to be "safe"
- Guess a sequence of prices $\{R_t\}_{t=1}^T$, with $R_T = R^\star$
- Solve the problem during the transition by **backward induction**
  - At T, the economy is back at its steady state (so we know the solution)
  - At $T-1$, the problem is:

$$
\begin{aligned}
V_{T-1}(a,y) = \max_{c_{T-1}, a_T} \quad & u(c_{T-1}) + \beta \mathbb{E} V^\star(a_T, y_T) \\
\text{s.t.} \quad & c_{T-1} + a_T = R_{T-1} a + y(1-\tau_{T-1}) \\
& \log y_T = \rho \log y + \sigma \epsilon_T \\
& a_T \geq -\phi \ \text{ (borrowing constraint)}
\end{aligned}
$$

Transitional Dynamics in Bewley Models - Algorithm

- Notice that we know the value function $V^{\star}(.)$, so we don't need to guess it.
  - We just need to solve for the policies $a_T(a, y)$ and $c_{T-1}(a, y)$
  - After solving for these policies, we can compute $V_{T-1}(a, y)$
- Repeat the same procedure for every $t \in [1, T-2]$
  - After this, we have $\{a_{t+1}(a, y)\}_{t=0}^{T}$ and $\{c_t(a, y)\}_{t=0}^{T}$, with $a_0(a, y) = a_T(a, y) = a^{\star}(a, y)$
- We can now use those policies to move **"forward"**
- That is, starting from $\Omega_0(.)$, update the firm distribution:

$$\Omega_{t+1}(A \times Y) = \int Q_t((a, y), A \times Y)\, \Omega_t(a, y)\, da\, dy$$

  - where $\Omega_0(.) = \Omega^{\star}(.)$ is known from our SS solution
  - For this step, we can use Young method to either approximate $Q_t(.)$ and $\Omega_{t+1}(.)$

Transitional Dynamics in Bewley Models - Algorithm

- Compute the period t aggregate asset demand:

$$\bar{a}_{t+1} = \int a_{t+1}(a, y)\Omega_t(a, y)\mathrm{d}a\mathrm{d}y$$

- For each $t \in [1, T-1]$, check that $\bar{a}_{t+1} = 0$. If not, update the guess for R_t accordingly
- Iterate until convergence of the full $\{R_t\}_{t=1}^{T-1}$ sequence